

Generation Interconnection System Impact Study Report

For Queue Project Q21-1660

St. John 138 kV
February 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1660. The Interconnection Customer proposes a 118.0 MW solar photovoltaic facility interconnecting at the St. John 138 kV substation in the Virginia area. The request was received on February 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 381 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,411,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Montano - Celeron 138 KV
Contingency: Celeron - Rio Dell 138 KV
Loading: 114.0% (Rating: 377 MVA)
DFAX: 48.3%
Pre-project loading: 98.9%
2. Facility: Montano Substation - Rio Dell 138 KV
Contingency: Celeron - Rio Dell Substation 138 KV
Loading: 113.8% (Rating: 381 MVA)
DFAX: 48.3%
Pre-project loading: 98.9%
3. Facility: Celeron - Rio Dell 138 KV line
Contingency: Montano - Rio Dell 138 KV L
Loading: 113.8% (Rating: 361 MVA)
DFAX: 45.9%
Pre-project loading: 98.9%
4. Facility: Palmyra - Glenbrook 138 KV
Contingency: Kearneyville Substation - Clark County Substation 345 KV Line
Loading: 111.0% (Rating: 362 MVA)
DFAX: 11.2%
Pre-project loading: 107.3%
5. Facility: Palmyra - Grimmway-Malaga 138 KV
Contingency: Kearneyville - Clark County 345 KV
Loading: 110.5% (Rating: 352 MVA)
DFAX: 11.2%
Pre-project loading: 106.7%
6. Facility: Clark County - Shelton 138 KV
Contingency: Artesia Country Club - Clark County 345 KV
Loading: 109.5% (Rating: 263 MVA)

DFAX: 5.8%

Pre-project loading: 106.9%

7. Facility: Killingsworth - Gallegos 138 KV

Contingency: Kearneyville - Clark County Substation 345 KV

Loading: 108.1% (Rating: 402 MVA)

DFAX: 11.2%

Pre-project loading: 104.8%

8. Facility: Millville - Sleepy Hollow Substation 138 KV

Contingency: Clark County - Kearneyville 345 KV

Loading: 101.0% (Rating: 530 MVA)

DFAX: 10.2%

Pre-project loading: 98.7%

9. Facility: Redman - Glenbrook 138 KV

Contingency: Kearneyville - Clark County 345 KV

Loading: 100.6% (Rating: 412 MVA)

DFAX: 11.2%

Pre-project loading: 97.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montano - Celeron 138 KV	Rebuild line with higher-capacity conductor	\$1,346,000	83% (\$1,115,000)
Montano Substation - Rio Dell 138 KV	Rebuild line with higher-capacity conductor	\$2,588,000	83% (\$2,143,000)
Rio Dell - Celeron 138 KV	Rebuild line with higher-capacity conductor	\$4,762,000	100% (\$4,762,000)
Palmyra - Glenbrook 138 KV	Rebuild line with higher-capacity conductor	\$1,113,000	31% (\$342,000)
Palmyra - Grimmway-Malaga 138 KV	Rebuild line with higher-capacity conductor	\$3,006,000	31% (\$924,000)
Clark County Substation - Shelton 138 KV	Reconductor line	\$390,000	13% (\$49,000)
Gallegos - Killingsworth 138 KV	Reconductor line	\$1,108,000	31% (\$341,000)
Millville - Sleepy Hollow Substation 138 KV	Reconductor line	\$1,890,000	23% (\$432,000)
Redman - Glenbrook 138 KV	Reconductor line	\$987,000	31% (\$303,000)
	Total allocated to project		\$10,411,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.